

Generation Interconnection Request System Impact Study Report

For Queue Project Q23-2271

Dysart 138 kV
December 30, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2271. The Interconnection Customer proposes a 58.0 MW solar photovoltaic facility interconnecting at the Dysart 138 kV substation in the West Virginia area. The request was received on January 01, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 734 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$102,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of International Paper Valliant Ok-Kearneyville 345 kV, the N Mesa-Sandalffoot 138 kV loads to 110.6% of its 425 MVA rating. Project DFAX: 7.5%.

Sandalffoot-Celeron 138 kV: 109.7% loading (442 MVA) following the outage of the International Paper Valliant Ok-Kearneyville 345 kV; DFAX = 7.5%.

For the loss of International Paper Valliant Ok-Kearneyville 345 kV L, the International Paper Valliant Ok-West Winchester 138 kV loads to 101.9% of its 636 MVA rating. Project DFAX: 8.3%.

The Shepherdstown-Lcec No. 5 138 kV is overloaded to 101.2% of its 582 MVA emergency rating for the loss of the Clark County-Kearneyville Substation 345 kV. The project contributes 8.5% DFAX (5.0 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N Mesa-Sandalffoot 138 kV	Rebuild line with higher-capacity conductor	\$2,133,000	2% (\$49,000)
Sandalffoot-Celeron 138 kV	Reconductor line	\$1,336,000	2% (\$31,000)
International Paper Valliant Ok-West Winchester 138 kV	Reconductor line	\$575,000	2% (\$11,000)
Shepherdstown-Lcec No. 5 138 kV	Reconductor line	\$768,000	1% (\$11,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$102,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.